

AN-001

AudioLab Bring-Up and Validation Guide

TLV320AIC3104 codec daughterboard with NUCLEO-F429ZI USB audio, SAI/I2S transport, I2C codec control, and STM32-owned DSP

Field	Value
Project	AudioLab
Document type	Application note / laboratory bring-up guide
Document ID	AN-001
Revision	Draft 0.1
Date	2026-06-16
Primary target	NUCLEO-F429ZI + TLV320AIC3104 stereo codec daughterboard
Status	Generated from repository specifications; hardware measurements and final firmware/host implementation evidence still required

Abstract

This application note gives a practical, end-to-end procedure for bringing up AudioLab as a laboratory USB-audio and embedded-DSP platform. It starts from safe power application, then validates VBUS/5V, 3V3, 1V8, codec reset, I2C control, SAI/I2S clocks, USB enumeration, playback, capture, DSP operation, diagnostics, and final demonstration evidence. The note intentionally treats presentation_es.pdf and the AudioLab reference folder as the source of truth for the implemented hardware. Any proof-of-concept material outside the TLV320AIC3104 AudioLab contract is not used as the system definition. [P1, R1-R5, S1-S6]

Key outcome

- A repeatable bring-up sequence that avoids enabling audio before power, reset, I2C, and clocks are known good.
- A validation matrix that maps observable lab evidence to system, hardware, firmware, power, and host-software requirements.
- A release checklist that clearly separates validated behavior from planned or unverified behavior.

Contents

1. Scope and operating assumptions
2. System overview
3. Hardware and power boundaries
4. Firmware operating model
5. Host-control and measurement workflow
6. Bring-up procedure
7. Validation and acceptance criteria
8. Troubleshooting guide
9. Release checklist
10. Source documents

1. Scope and operating assumptions

Source-truth reconciliation: this revision was checked against the current AudioLab Spanish presentation and local hardware reference folder. The hardware source of truth is the TLV320AIC3104 AudioLab board, with exported schematic/PCB/Gerber/DRC evidence; firmware migration to TLV320AIC3104, USB Audio, SAI/I2S, and STM32-owned DSP remains pending, and PCB fabrication/assembly plus rail/audio measurements still must be completed before validated-operation claims. [P1]

AudioLab is a NUCLEO-F429ZI-based USB audio and embedded DSP platform using the TLV320AIC3104 stereo audio codec. The repository separates hardware design, STM32 firmware, PC host software, specifications, tests, and tools. The committed hardware is the most complete subsystem and includes Altium source, generated outputs, exported schematics/PCB review artifacts, a STEP model, and board render material. [S1]

In scope

- Safe laboratory bring-up of the codec daughterboard with the NUCLEO-F429ZI target.
- Verification of the implemented TLV320AIC3104 power rails: VBUS/5V input, TLV75533-generated 3V3 for AVDD/DRVDD/IOVDD, TLV70018-generated 1V8 for DVDD, reset behavior, I2C control, and SAI/I2S audio.
- Validation of 48 kHz stereo playback or capture, at least 16-bit PCM samples, and at least one real-time DSP path.
- Use of a PC host over USB for audio streaming plus a host-control or diagnostic path once the firmware contract is available.

Out of scope

- Consumer-product certification, medical or safety-critical operation, and production-ready commercial claims.
- Custom host operating-system audio drivers; the preferred model is to use the normal PC audio stack for audio and a separate documented control path for configuration and diagnostics.
- Any claim that the USB-C power/data path is fully validated before VBUS/5V, 3V3, 1V8, USB_D_P/USB_D_N protection, reset, and startup behavior have been measured.

Status conventions used in this note

Label	Meaning
Required	A project requirement or bring-up step that must be satisfied before claiming the corresponding behavior.
Recommended	A preferred implementation or measurement that improves confidence but may be deferred with justification.
Open	A choice or measurement that must be closed before release or presentation as complete.
Do not claim	A behavior that must not be advertised until evidence exists in the validation record.

2. System overview

The system boundary is a PC host connected through AudioLab USB-C/VBUS and protected USB_D_P/USB_D_N lines to the STM32F429ZI target on the NUCLEO-F429ZI board. The STM32 controls the TLV320AIC3104 over I2C, exchanges PCM audio with it over an SAI/I2S-compatible serial interface,

performs buffering and real-time DSP, and exposes enough state to diagnose USB, codec, streaming, and buffer faults. [P1, S2]

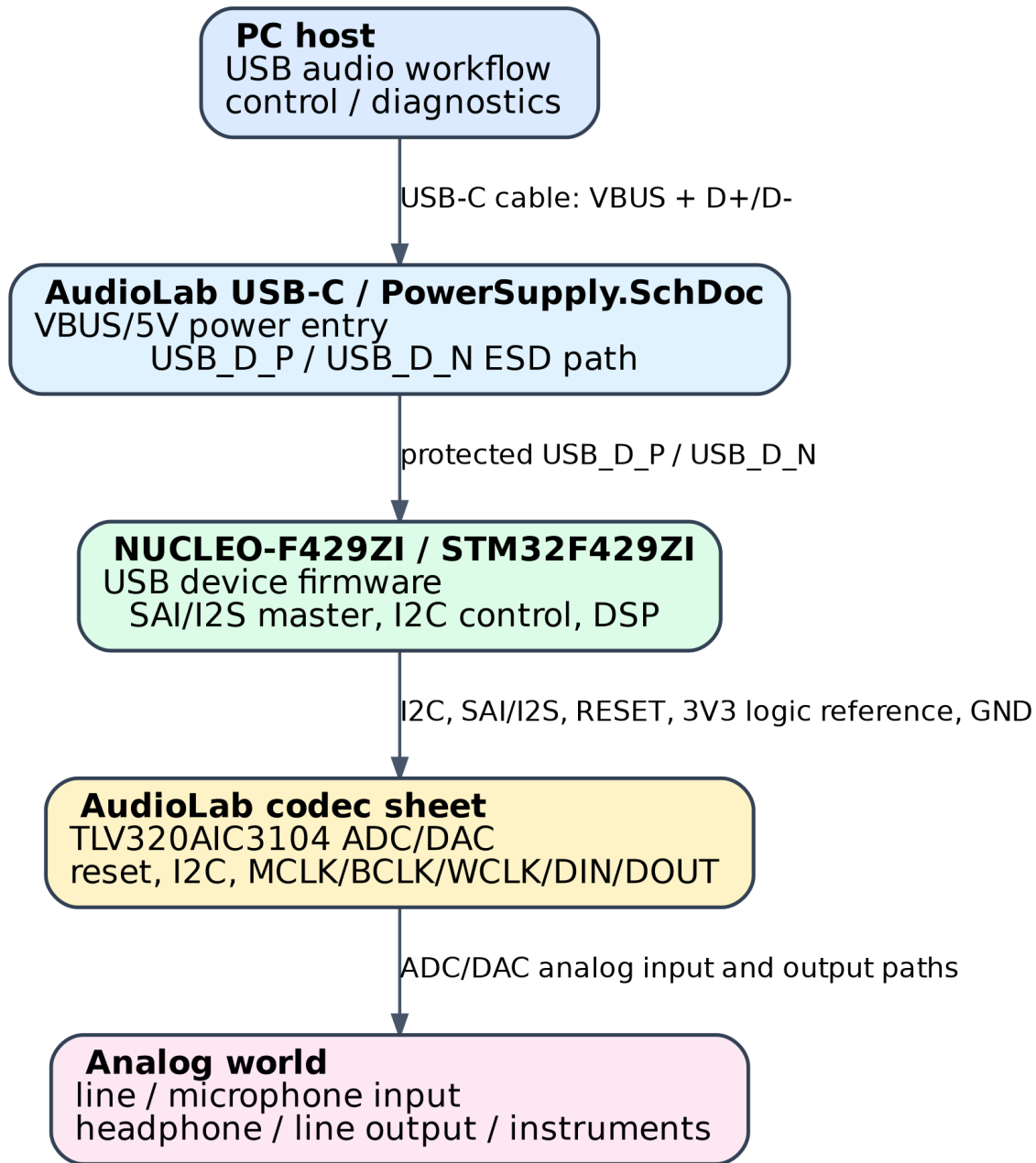


Figure 1. AudioLab system boundary.

Subsystem	Main responsibility	Evidence expected during bring-up
PC host	USB audio workflow, configuration, diagnostics, validation scripts	Device enumeration, status output, command responses, audio playback/capture records
STM32 firmware	USB device behavior, codec control, SAI/I2S transport, DMA buffering, DSP, faults	State machine status, I2C result, clock status, underrun/overflow counters, DSP active flag
Codec daughterboard	TLV320AIC3104 power, reset, analog routing, USB-C/VBUS power entry, USB data	Measured VBUS/5V, 3V3, 1V8, reset waveform, I2C activity, SAI/I2S clocks, analog

	protection, measurement access	input/output waveforms
Lab instruments	Independent measurement of power, clocks, and audio signals	DMM/scope/analyzer screenshots, current logs, loopback measurement notes

3. Hardware and power boundaries

The AudioLab daughterboard is a codec-centered mixed-signal board. Its architecture includes the TLV320AIC3104 codec, board-to-board connections to the NUCLEO-F429ZI headers, codec power/reset/control, SAI/I2S audio, line or microphone input evaluation, headphone or line output evaluation, and measurement access. [S3]

3.1 Digital interface baseline

Function	Direction	Project net / controller side	Codec signal	Bring-up check
Master clock	STM32 -> codec	CODEC_MCLK from STM32 SAI/I2S master	MCLK	frequency present before audio claim
Bit clock	STM32 -> codec	CODEC_BCLK from STM32 SAI/I2S master	BCLK	correct relation to sample format
Word clock	STM32 -> codec	CODEC_WCLK from STM32 SAI/I2S master	WCLK	48 kHz for baseline mode
Playback data	STM32 -> codec	CODEC_DIN from STM32 playback path	DIN	changes with playback stimulus
Capture data	codec -> STM32	CODEC_DOUT to STM32 capture path	DOUT	changes with input/loopback signal
I2C clock	STM32 -> codec	CODEC_I2C_SCL from STM32 I2C controller	SCL	activity during init/readback
I2C data	bidirectional	CODEC_I2C_SDA bidirectional I2C data	SDA	ACK/readback observed

Codec signal names are from the codec perspective: STM32 playback data is written to codec DIN, and codec capture data is read from codec DOUT. The preferred initial clocking model is STM32 master and codec slave at 48 kHz stereo, at least 16-bit PCM, with 32-bit slots preferred for alignment and possible later 24-bit operation. [S3]

3.2 Power architecture

The implemented power architecture uses a USB-C/VBUS power entry, then a protected 5V path with fuse, filtering, and decoupling. TLV75533 generates 3V3 for the TLV320AIC3104 AVDD, DRVDD, and IOVDD

domains; TLV70018 generates 1V8 for DVDD. USB_D_P and USB_D_N pass through protection before reaching the STM32 system. [P1, R3-R5]

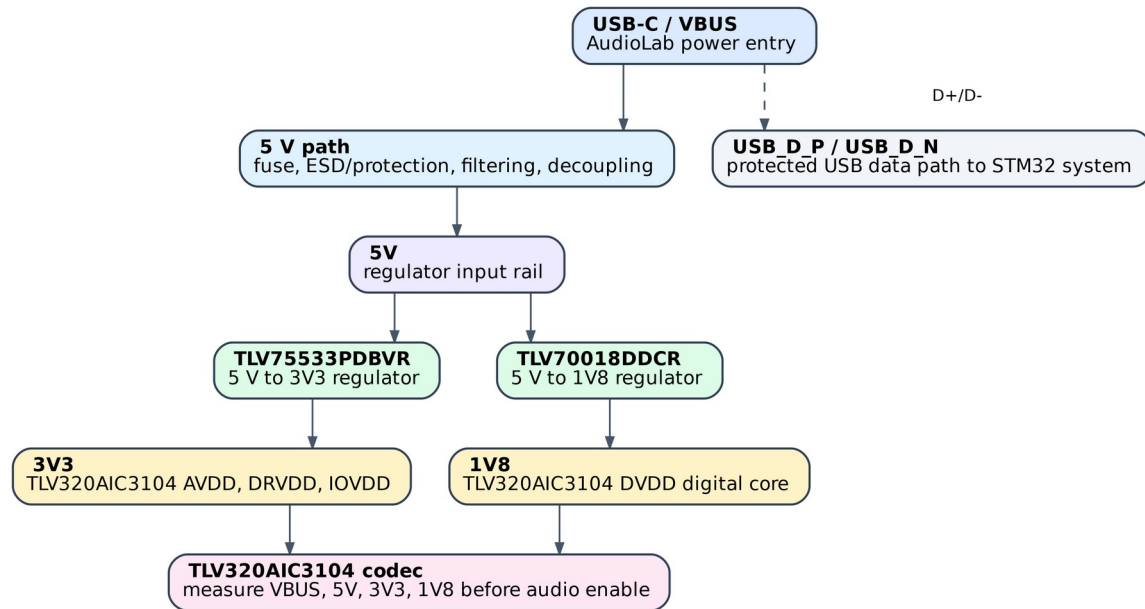


Figure 2. Implemented AudioLab USB-C/VBUS and codec power tree.

Rail	Nominal	Loads	First measurement
VBUS	USB nominal 5 V	USB-C power entry before local 5V path	verify connector input, polarity, current limit, and ESD/protection path
5V	5.0 V nominal	input to TLV75533 and TLV70018 regulators after fuse/filter/decoupling	verify before downstream regulator checks
3V3	3.3 V	TLV320AIC3104 AVDD, DRVDD, IOVDD	verify codec analog/output-driver and digital-I/O rail before reset release
1V8	1.8 V	TLV320AIC3104 DVDD	verify digital-core rail before I2C/audio streaming

3.3 Power-up safety rule

Do not enable normal audio operation until VBUS/5V, 3V3, and 1V8 have been measured, reset behavior is known, and I2C access succeeds. The safe startup rule is: apply current-limited VBUS/5V, verify 3V3 and 1V8, hold codec RESET inactive until rails are stable, release RESET under firmware control, initialize the codec over I2C, and unmute enabled audio paths only after the firmware audio path is ready. [P1, R1-R4, S4]

4. Firmware operating model

The AudioLab firmware runs on the STM32F429ZI and owns startup, clock/peripheral configuration, USB Audio device behavior, optional host control, TLV320AIC3104 reset and I2C register control, SAI/I2S audio transport, DMA/interrupt handling, playback/capture/full-duplex buffering, real-time DSP, diagnostics, and fault reporting. [S5]

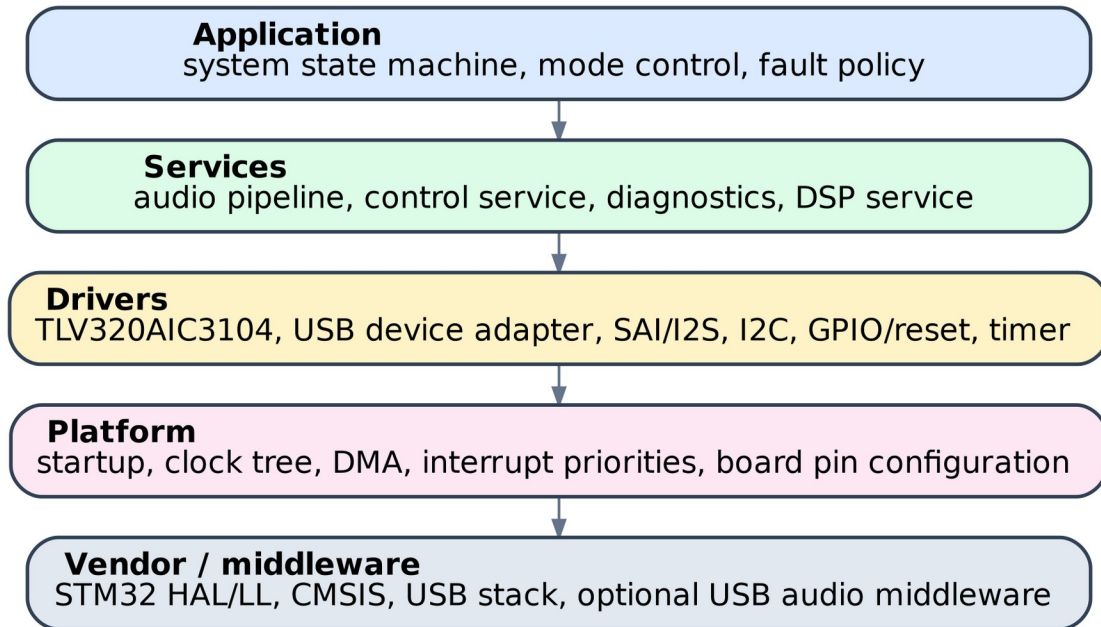


Figure 3. Recommended firmware layering.

Layer/module	Must own	Must not own
Codec driver	TLV320AIC3104 reset, register access, named codec modes, mute safety	Scattered raw register writes from application code
Audio transport	SAI/I2S, DMA movement, clock master behavior, half/full-transfer events	Long DSP work in interrupt context
USB audio	USB enumeration and streaming endpoints	DSP algorithm decisions
Host control	Command parsing, validation, dispatch, explicit status	Direct codec/DSP changes from USB interrupt path
DSP service	Gain/mute/EQ/FIR/compressor/meter blocks, state and parameter validation	Heap allocation in the audio callback path
Diagnostics	USB, codec, stream, DMA, I2C, buffer, DSP and fault counters	Hiding failures behind silent fallback

4.1 Baseline firmware mode

Property	Baseline
Sample rate	48 kHz
Channels	stereo
PCM sample width	16-bit minimum
Audio slot width	32-bit preferred
Codec role	SAI/I2S slave
STM32 role	SAI/I2S clock master
USB role	USB device
Codec control	I2C

Audio transport	DMA-backed SAI/I2S
Scheduler	bare-metal main loop with interrupt/DMA callbacks for first implementation

4.2 Runtime state machine

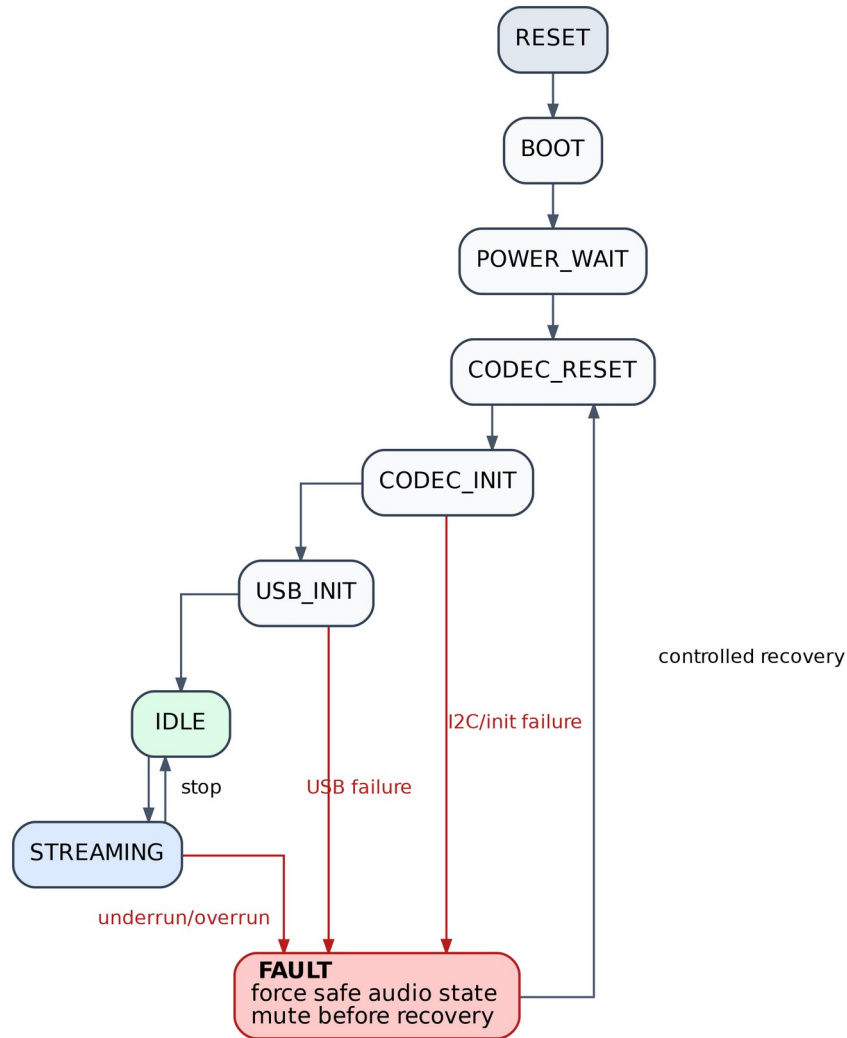


Figure 4. AudioLab firmware state machine and safe fault path.

Each state shall define legal commands. Illegal commands must return explicit errors. Any fault that can leave audio outputs unstable shall force a safe state before recovery is attempted. [S5]

5. Host-control and measurement workflow

The host-side software configures, observes, and validates the device; it does not own real-time audio transport, codec register timing, DMA, or hardware fault recovery. The first host-side implementation should be a command-line utility plus a small library. A GUI should wait until CLI workflows and the protocol are stable. [S6]

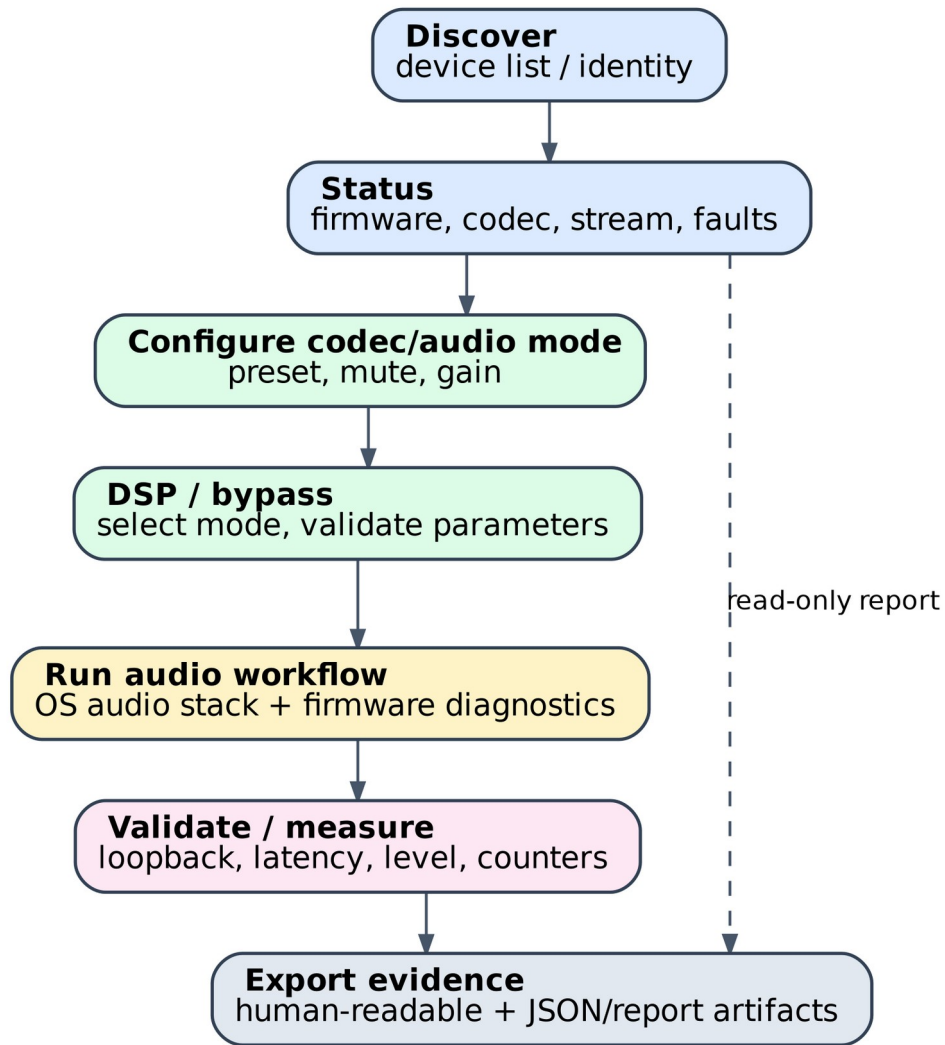


Figure 5. Recommended host-control and measurement workflow.

Command group	Purpose	Expected safe behavior
audiolab device list / info	Find and identify devices	No state change; explicit no-device or multiple-device errors
audiolab status	Read firmware, codec, stream, DSP, and fault state	Read-only; marks stale or unknown state clearly
audiolab codec preset <name>	Select validated codec routing mode	Validate preset and state; mute before unsafe transitions when supported
audiolab dsp set / param	Select DSP block and parameters	Range-check parameters before sending; apply at safe block boundary
audiolab diagnostics dump	Collect counters and last fault	Structured output for reports and tests
audiolab validate / measure	Run repeatable validation or measurement workflow	Records method, input data, output files, and pass/fail result

6. Bring-up procedure

Use the sequence below in order. The ordering matters because later checks depend on earlier evidence: audio streaming must not be claimed before power rails, reset behavior, I2C control, and digital clocks are known good.

Step	Gate	Action	Required evidence
------	------	--------	-------------------

0	Documentation and visual inspection	Confirm schematic/PCB review status, inspect for shorts, missing parts, orientation, connector pin-1, solder bridges, and unpopulated DNP decisions.	Inspection notes; photos; known assembly deviations.
1	Current-limited first power	Power VBUS/5V through an approved current-limited source. Treat USB-C VBUS as the AudioLab power entry and do not proceed because USB data appears connected.	VBUS and 5V voltage/current; no hot components; no unexpected rail shorts.
2	Rail measurement	Measure VBUS, 5V, 3V3, 1V8, ground references, and idle current before enabling audio.	Rail table with DMM readings and current limits.
3	Reset behavior	Verify codec RESET remains asserted until rails are stable and firmware intentionally releases it.	Scope capture or timestamped measurement of RESET relative to rails.
4	I2C control	Run a repeatable read/write or identification procedure through the codec driver.	SCL/SDA scope capture, ACK/readback log, explicit error behavior on failure.
5	SAI/I2S clocks	Enable baseline 48 kHz stereo clocking with STM32 as SAI/I2S master and codec as slave.	MCLK, BCLK, WCLK frequencies and expected relationships.
6	Playback path	Play a known digital stimulus from the PC through USB and firmware to codec DAC/output.	Analog output waveform, level, no underrun counter growth, stimulus file name.
7	Capture path	Inject a known analog signal or loopback output into input; capture through codec ADC and USB.	Captured file/spectrum/level, DOUT activity, overrun counter behavior.
8	DSP mode	Enable at least one observable DSP block, such as gain, mute, EQ, FIR, compressor/limiter, or level meter.	Before/after measurement and parameter log; bypass comparison.
9	Host diagnostics	Use host-control path to read device identity, state, counters, codec mode, stream state, DSP mode, and last fault.	Human-readable and structured diagnostic output.
10	Fault handling	Exercise unsupported commands, malformed inputs, disconnected-device cases, and controlled fault	Explicit errors; no corrupted codec state; audio returns to safe state.

		clear where supported.	
11	Final demo run	Run the chosen acceptance profile without manual hidden steps.	Pass/fail record, measurement artifacts, known limitations.

7. Validation and acceptance criteria

The minimum acceptable demonstration shows USB enumeration, successful TLV320AIC3104 initialization over I2C, valid MCLK/BCLK/WCLK digital audio clocks, 48 kHz stereo playback or capture, one real-time DSP operation, visible or logged system status, measured or observed evidence for the active audio path, and no known unsafe power or output state in intended laboratory use. [P1, S2]

Requirement area	Acceptance evidence	Do not claim until
USB audio	Device enumerates on at least one PC; audio workflow starts reproducibly	Enumeration and host behavior are logged
Codec control	I2C init succeeds through codec driver; register errors are reported	A repeatable read/write or identification procedure passes
SAI/I2S	MCLK/BCLK/WCLK/data signals are measured or diagnosed	Clock and data direction are confirmed against schematic
Playback	Known stimulus is observed at analog output	Output level and underrun behavior are recorded
Capture	Known input or loopback is captured to host	Captured data and overrun behavior are recorded
DSP	Bypass and active DSP differ in a measurable or visible way	Parameters and result are documented
Power safety	Rail voltages/currents recorded at no-load, idle, and active modes	No unsafe rail or output behavior is known
Host control	Commands return explicit success/error and structured output	Malformed/disconnected cases are tested

7.1 Measurement record template

Item	Value / file / observation
Board revision and assembly notes	
Firmware build ID	
Host software version / commit	
Power source and current limit	
Rail readings: VBUS, 5V, 3V3, 1V8	
Codec I2C readback result	
Clock measurements: MCLK, BCLK, WCLK	
Playback stimulus and output evidence	
Capture input and captured file	
DSP mode, parameters, and before/after evidence	
Diagnostics output path	
Fault tests performed	
Known limitations	

8. Troubleshooting guide

Symptom	Likely checks	Safe next action
No codec I2C ACK	Verify 3V3, 1V8, I2C pull-ups,	Keep outputs muted; run bus scan only if safe; scope SCL/SDA

	reset release, SCL/SDA mapping, address/page handling	
I2C works but no audio clocks	Check STM32 clock tree, SAI pin alternate functions, firmware state, DMA init	Return to IDLE; enable clocks without streaming and measure MCLK/BCLK/WCLK
WCLK not 48 kHz	Check PLL/audio clock config, slot/frame format, SAI master settings	Do not claim baseline mode; log measured clock and firmware config
Playback underruns	Check USB packet timing, ring-buffer fill, DMA callbacks, ISR load, DSP cost	Enable bypass; increase buffer instrumentation before changing algorithm
Capture overruns	Check USB IN path, capture ring buffer, host read behavior, DMA RX event handling	Record counter growth and capture-only test before full duplex
Analog output pop/noise at startup	Check reset/mute sequence, output coupling, rail ramp, codec output mode	Keep outputs disconnected or muted until reset and init sequence is fixed
Host command appears to succeed but device state unchanged	Check response status, protocol version, command legality in current state	Treat byte write as insufficient; require explicit firmware success status
Power rail droop during streaming	Check TLV75533/TLV70018 regulator margin, headphone load, analog output mode, current limit, VBUS/5V source	Stop streaming; measure rail current in idle, playback, capture, and DSP modes

9. Release checklist

- ☐ Repository documentation matches the current folder structure and hardware source of truth.
- ☐ Hardware schematic/PCB evidence is current, including exported schematics, PCB files, Gerbers/drills, and DRC status.
- ☐ VBUS, 5V, 3V3, and 1V8 have measured values and current records before normal audio operation.
- ☐ I2C codec initialization is repeatable and failures are reported explicitly.
- ☐ SAI/I2S clocks and data direction are measured or diagnosed before full audio streaming is claimed.
- ☐ USB enumeration is validated on at least one PC host.
- ☐ Playback is validated with a repeatable audio stimulus.
- ☐ Capture is validated with a repeatable input or loopback signal.
- ☐ At least one DSP mode is validated with an observable or measurable effect.
- ☐ Host-control malformed-input, unsupported-command, and disconnected-device cases are tested where the host-control path exists.
- ☐ Unsupported or unvalidated requirements are explicitly marked before the project is presented as complete.
- ☐ No known unsafe power or audio-output state remains under intended laboratory use.

10. Source documents

The note was compiled from the AudioLab repository documentation available in the Google Drive folder. Bracketed source labels are used for traceability within this application note.

Source	Document	Used for
P1	docs/presentations/audiolab/presentation_es.pdf	Primary source-truth check for implemented hardware, power tree, current status, validation

		plan, and project claims.
R1	hardware/references/ tlv320aic3104.pdf	Codec capabilities, supply-domain ranges, I2C control bus, audio serial formats, ADC/DAC and headphone/line behavior.
R2	hardware/references/nucleo-f429zi.pdf and UM1974 Nucleo-144 manual	NUCLEO-F429ZI / STM32 Nucleo-144 board context, USB/device and expansion-board assumptions.
R3	hardware/references/tlv755p.pdf	3.3 V regulator reference for the implemented 5V-to-3V3 rail.
R4	hardware/references/tlv700.pdf	1.8 V regulator reference for the implemented 5V-to-1V8 rail.
R5	hardware/references/usb4105-spec.pdf	USB-C receptacle reference for the implemented USB-C/VBUS entry.
S1	README.md	Repository purpose, repository boundaries, hardware status, and source organization.
S2	docs/specs/system/ requirements.md	System context, functional requirements, performance baseline, validation requirements, acceptance criteria.
S3	docs/specs/hardware/ architecture.md	Hardware boundary, TLV320AIC3104 daughterboard scope, signal mapping, measurement access.
S4	docs/specs/power/ architecture.md	Power safety intent and validation order; reconciled here to the implemented rails in presentation_es.pdf.
S5	docs/specs/firmware/ architecture.md	Firmware responsibilities, layering, state machine, buffering, DSP and diagnostics policy.
S6	docs/specs/software/ architecture.md	Host software scope, CLI-first model, protocol boundary, diagnostics, measurement workflow.

Appendix A. Suggested lab evidence package

- Board photos: top, bottom, connector orientation, relevant rework.
- Power evidence: current-limited first power, VBUS/5V/3V3/1V8 readings, idle/playback/capture/DSP current.
- Digital evidence: I2C transaction capture, MCLK/BCLK/WCLK screenshots, DIN/DOOUT observations.
- Audio evidence: playback stimulus, output waveform/spectrum, capture file, loopback path note.
- Firmware evidence: build ID, state-machine log, diagnostics dump, fault counter snapshot.
- Host evidence: command transcript, structured JSON output when available, validation profile result.
- Limitations: planned behavior not yet validated, open hardware decisions, unresolved anomalies.